



SERINA SAKHARE

Aspiring FPGA / Hardware Engineer | ECE Undergraduate

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PROFILE SUMMARY

Electronics & Communication Engineering undergraduate with hands-on FPGA/RTL design and hardware interfacing experience across two internships. Strong hardware fundamentals in Verilog/VHDL RTL design, clock management, DDS-based sine-wave synthesis, FIFO buffering, and DAC interfacing, with hardware debugging using ILA (Integrated Logic Analyser) on Xilinx Vivado. Additional exposure to analog circuit design in Cadence Virtuoso, network-scheduling simulation in NS-3, and IoT embedded systems. Comfortable in Windows subsystem for Linux with working knowledge of Python, C/C++.

TECHNICAL EXPERIENCE

Summer Intern — FPGA Costas Loop for Carrier Synchronisation — Centum Electronics

Jun – Jul 2026

- Developed a Costas Loop based BPSK carrier recovery system by modelling and validating the algorithm in MATLAB/Simulink.
- Translating the design into VHDL RTL, implementing components such as the NCO, I/Q mixers, LPF, and PI loop filter.
- Building an FPGA digital communication design.

Summer Intern, Navigation Receiver Division — (ISRO)

May – Jul 2025

- Designed and implemented a BPSK Modulator on a Kintex-7 FPGA (Verilog HDL, Xilinx Vivado), building a DDS-based sine-wave generator core, FIFO buffering IP for data-rate matching, and DAC interfacing (FMC150 daughtercard).
- Configured MMCM IP Cores for on-chip clock synthesis and interfaced with a CDCE external clock-generator device for receiver reference clocking; used ILA to debug timing issues directly on hardware.
- Applied basic Tcl scripting in the Vivado Tcl console to resolve timing constraint violations.
- Verified hardware validation results on oscilloscope and spectrum analyser.

Winter Intern — Human Spaceflight and Advanced Technology Area - (ISRO)

Dec 2025 – Jan 2026

- Designed a two-stage operational amplifier in 65nm CMOS using foundry PDKs in Cadence Virtuoso at schematic level.
- Implemented current mirrors, differential pair, common source gain stage, and source follower buffer, optimizing device sizing and biasing for performance trade-offs.
- Performed AC, transient, gain/phase margin, and unity-gain bandwidth analysis to evaluate stability and frequency response.
- Achieved 35 dB open loop gain and 2.72 GHz UGB, validating phase margin through frequency sweep simulations.

RELEVANT PROJECTS

Queue Management for QoS Enhancement — Independent Project

Feb – Apr 2026

- Simulated network queuing and scheduling in NS-3, implementing FIFO and Weighted Round Robin (WRR) scheduling to manage congestion across nodes under varying load.
- Designed a priority-based queueing to enhance QoS under congestion across varying traffic loads.
- Analysed throughput, latency, packet delivery ratio to optimize resource allocation.

Smart Healthcare Monitor System — Independent Project

Sep – Oct 2025

- Built an IoT-based real-time health monitoring system on an ESP32 microcontroller along with DHT11 sensor for temperature and humidity, real-time SpO2 and heart-rate measurement.
- Integrated the Blynk IoT platform for wireless streaming and remote visualization of vitals (SpO2, heart rate, temperature, humidity) on a mobile dashboard.

EDUCATION

B.Tech, Electronics & Communication Engineering — Manipal Institute of Technology (2023 – 2027)

CORE SKILLS & TOOLS

FPGA / Hardware: Verilog, VHDL, RTL Design & Testbenches, Clocking (MMCM, CDCE ICs), DDS, FIFO, DAC Interfacing, IP Core Integration

Programming: Python, C, C++, Basic Tcl Scripting

Tools: Xilinx Vivado, Cadence Virtuoso, MATLAB / Simulink, NS-3, Arduino IDE

CERTIFICATIONS & LANGUAGES

India Semiconductor Workforce Development Program — IISc | Synopsys | Samsung (Fellowship Grade II)

LEADERSHIP & COLLABORATION

- Treasurer, Economics & Finance Society of Manipal — managed budgeting and financial planning for a 40+ member society.
- Vice President, Aaina Dramatics Club — led a 40+ member team, coordinating cross-functional planning and representing the club at national-level competitions (BITS Goa, IIM Bangalore).